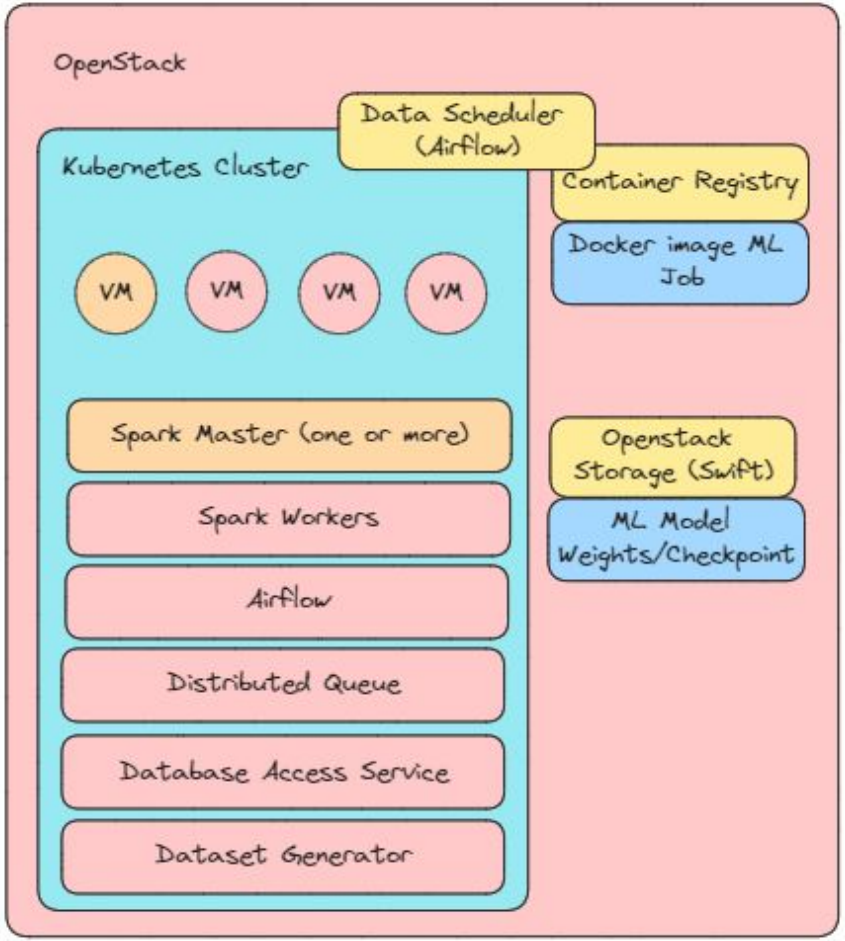
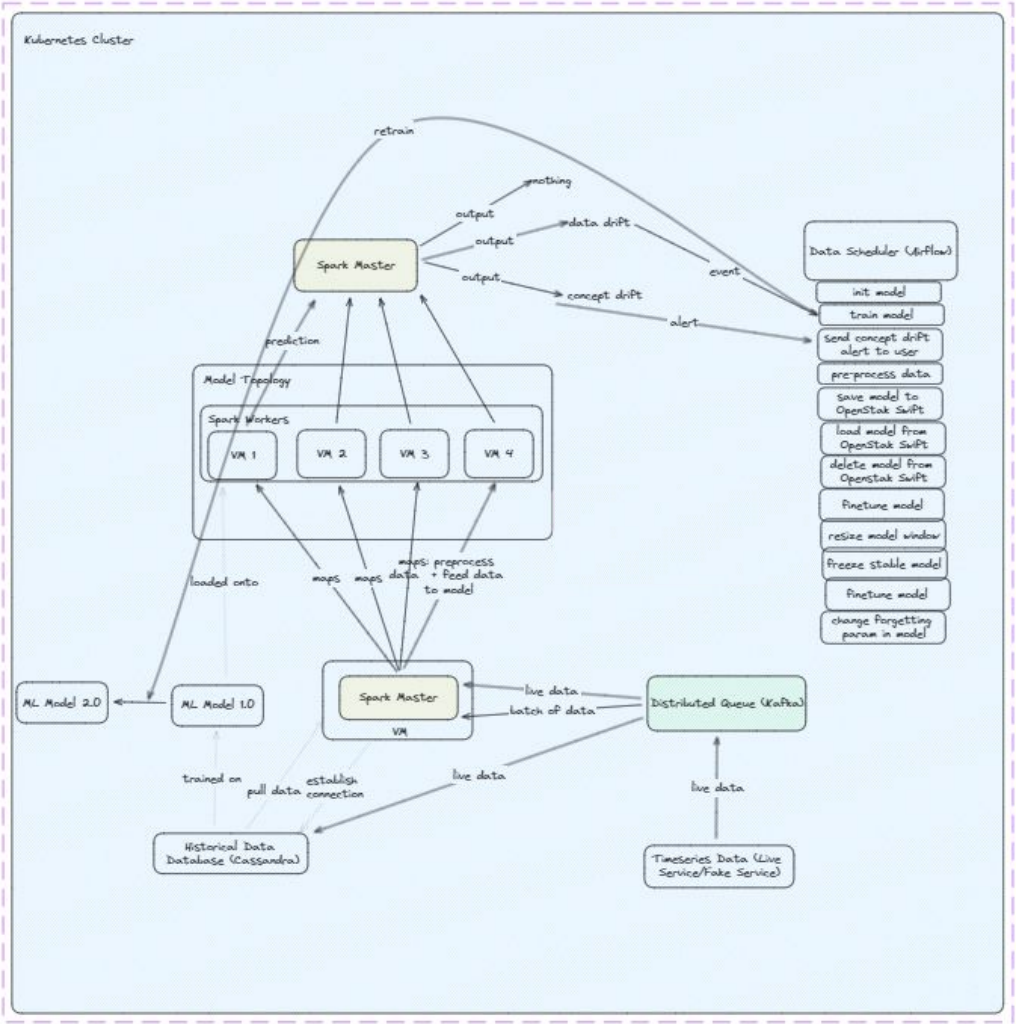
Data and Concept Drift Detection

Group 6

High Level View: Platforms/CICD



In-depth view: Services



Dataset Generation

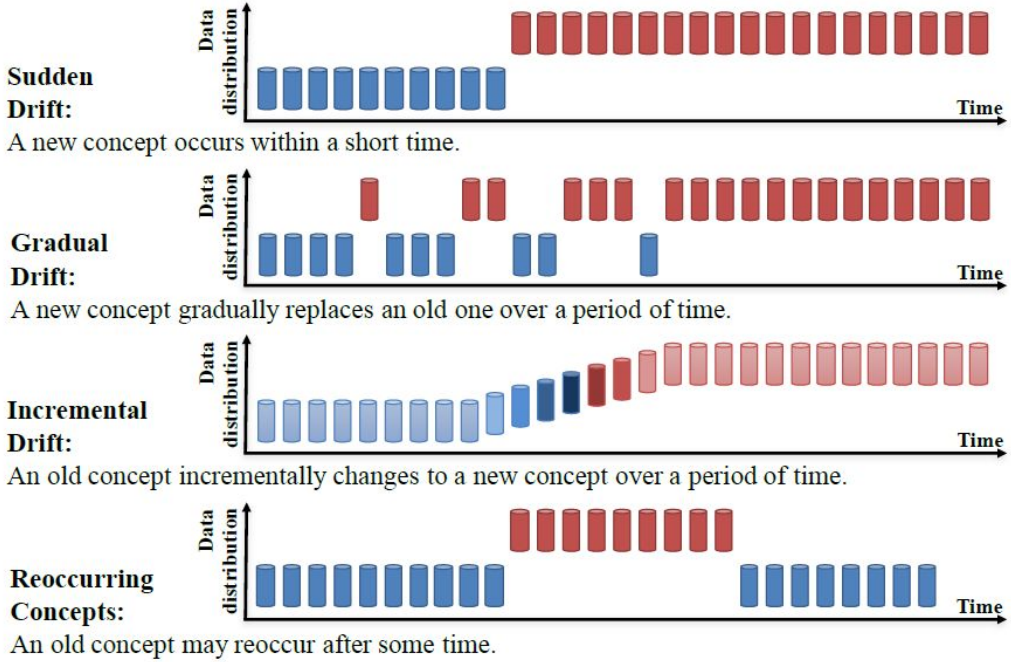


Fig. 4. An example of concept drift types.

Airflow Model Tasks

1. Time of drift (When)

=> **1 Action** : event = determine **duration** of time drift => resize the window for the model algorithm

2. Event for Severity of drift (How)

=> **2 Actions** : event = severity drift value returned by

- Low severity drift => incremental learning of the current model
- High severity drift =? Discard current model, retrain

3. The drift regions of concept drift (Where)

1. Training new models for global drift: **two learners: the stable learner and the reactive learner.**

   => 2 **Actions** : event = drift region detected

   - Replace stable kdqtree with RNN => the RNN can be finetuned on the new data and adapt to it
   - Change the forgetting parameter to a higher value for the models
2. Model ensemble for recurring drift: employ kdqtree and RNN
3. Adjusting existing models for regional drift: use RNN

References